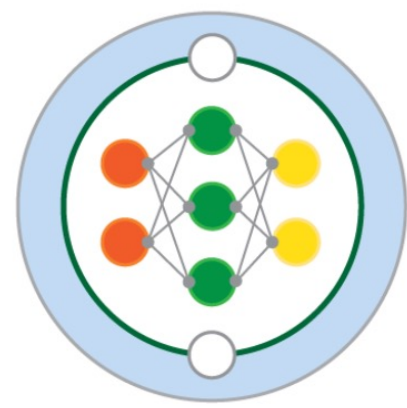


Engineering Convolutional Neural Networks to Enhance Climate Models

Priscila Madrid¹, Griffin Mooers², and Paul O’Gorman²

¹Department of Computer Science, University of Texas at El Paso ²Department of Earth, Atmospheric, and Planetary Science, Massachusetts Institute of Technology



THE UNIVERSITY OF TEXAS
AT EL PASO

Introduction

- Conventional approaches such as General Circulation Models (GCMs) have made significant strides in providing a more accurate representation of clouds, rainfall, and climate feedbacks but are constrained by imprecise parameterizations (representations) as well as **computational limits**.



Figure 1: Image of the Bridges-2 supercomputer.

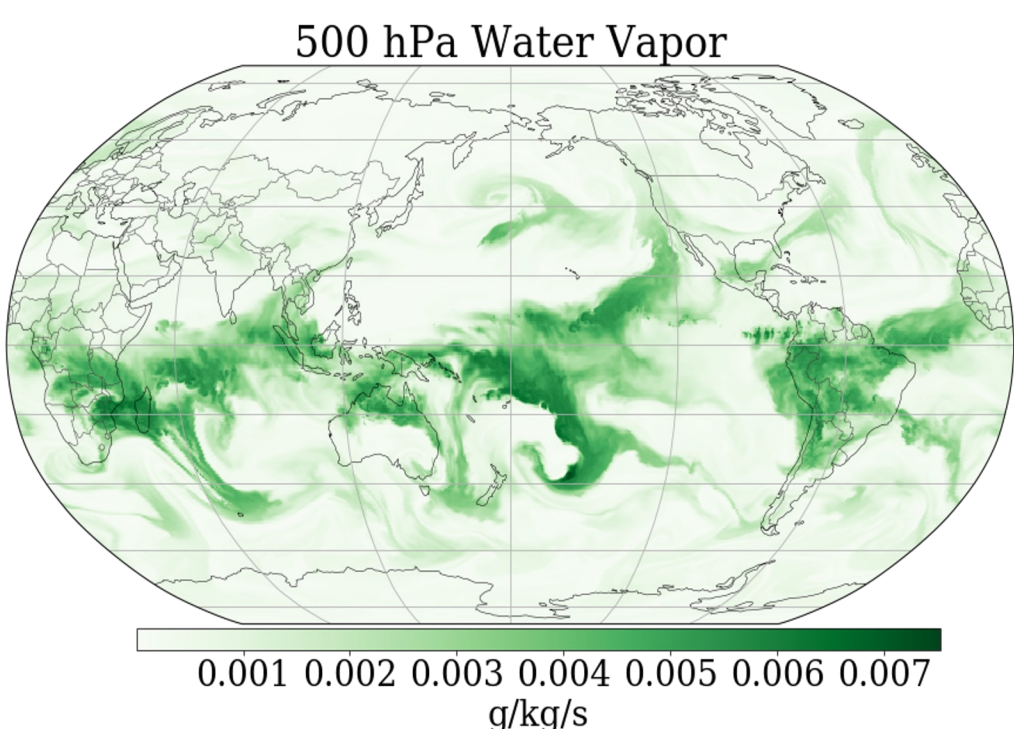
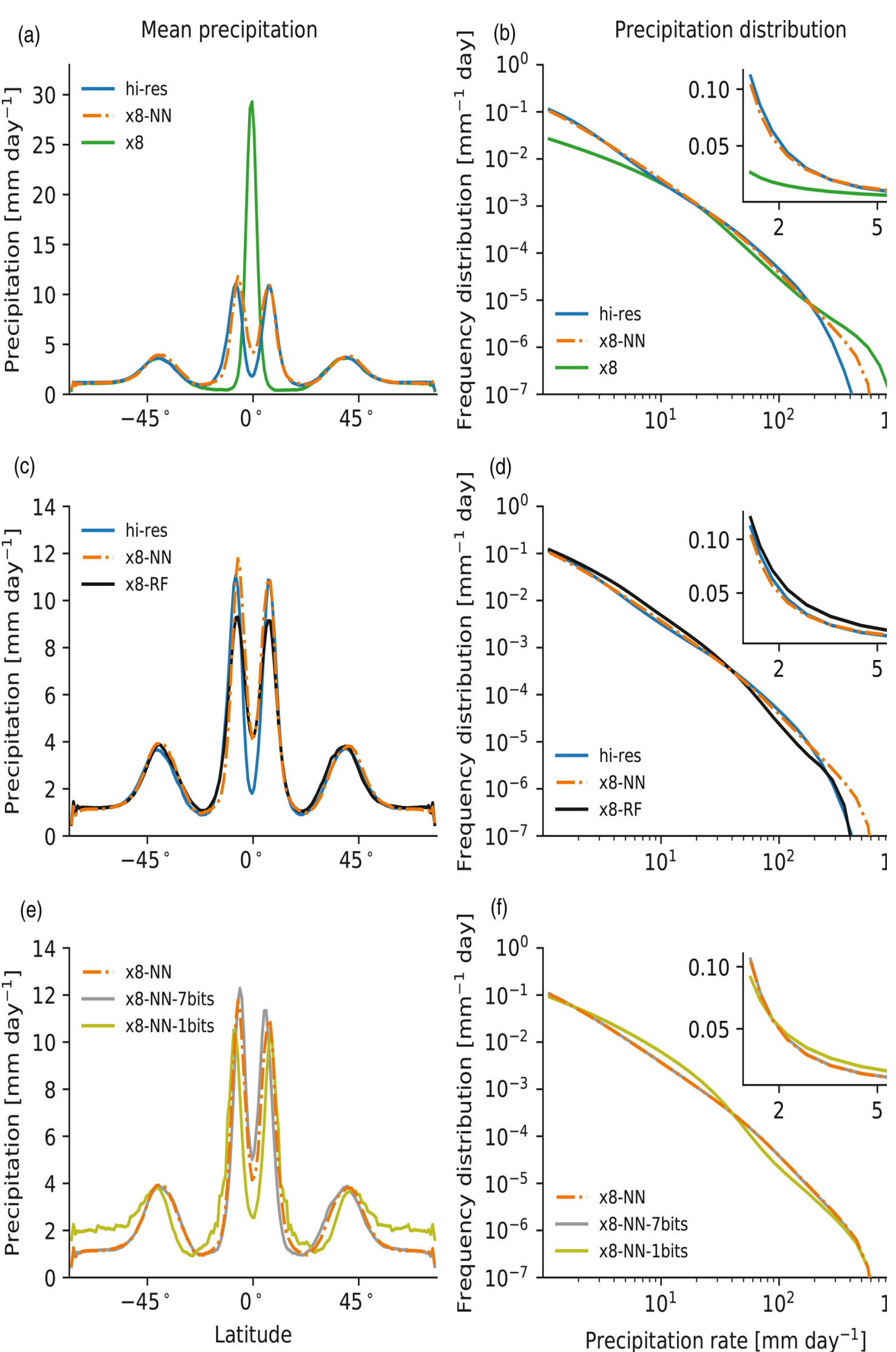


Figure 2: Example of high-resolution snapshot data.

- These physically based approaches may **struggle** with the representation of **small-scale convection** (Randall, 2003).

- An alternative to traditional parameterizations is to develop new subgrid parameterizations using **machine learning** (ML) algorithms (Yuval & O’Gorman, 2020).

Project So Far



Graph 1: Geophysical Research Letters, Volume: 48, Issue: 6, First published: 06 February 2021, DOI: (10.1029/2020GL091363)

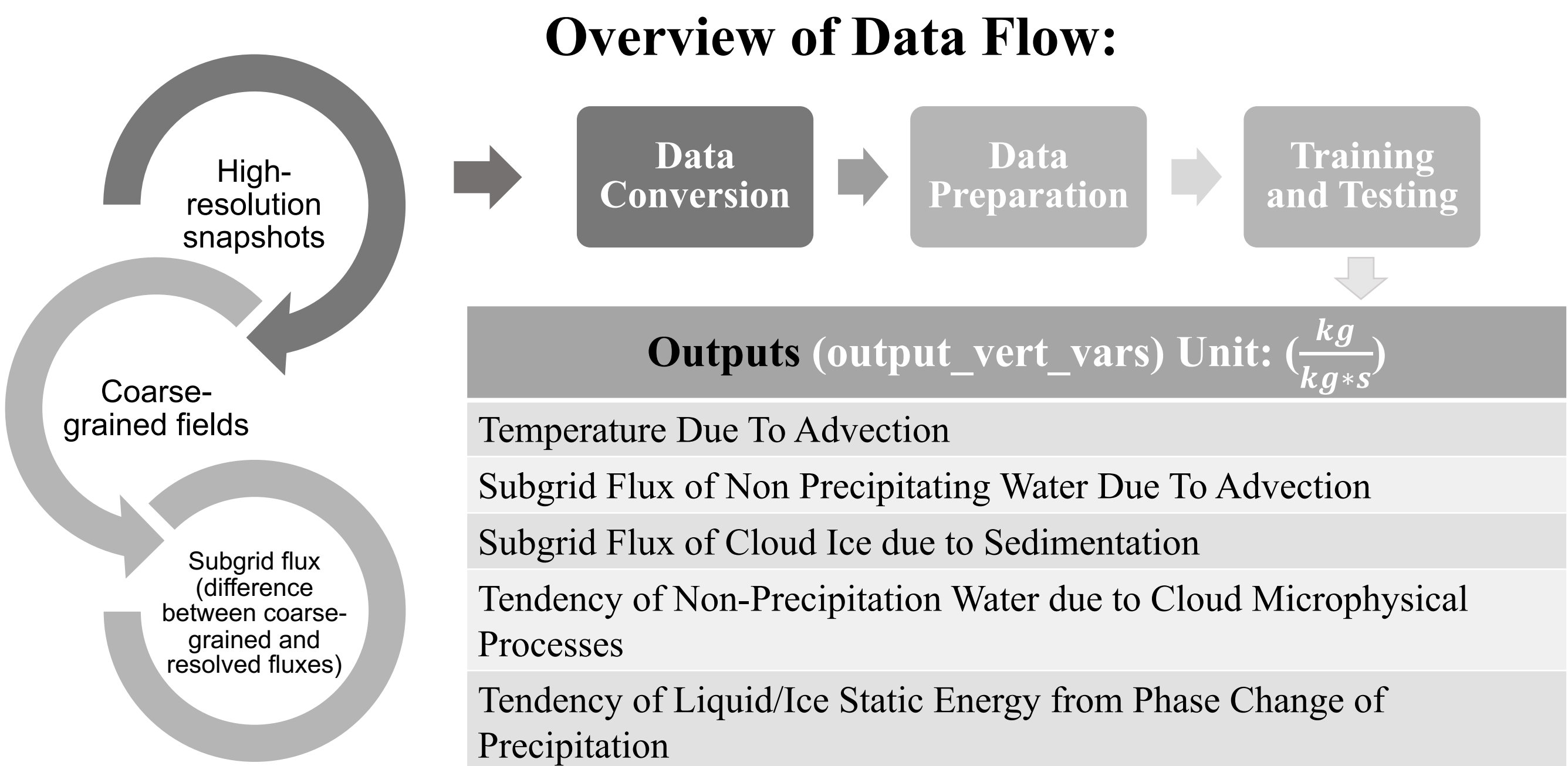
- A neural network (NN) for parameterization was developed using high-resolution atmospheric simulation data, ensuring that physical constraints were respected by predicting subgrid fluxes rather than tendencies.
- The NN parameterization produced stable simulations that closely replicated the high-resolution climate, showing similar accuracy to a successful random-forest parameterization while requiring less memory.

Main question

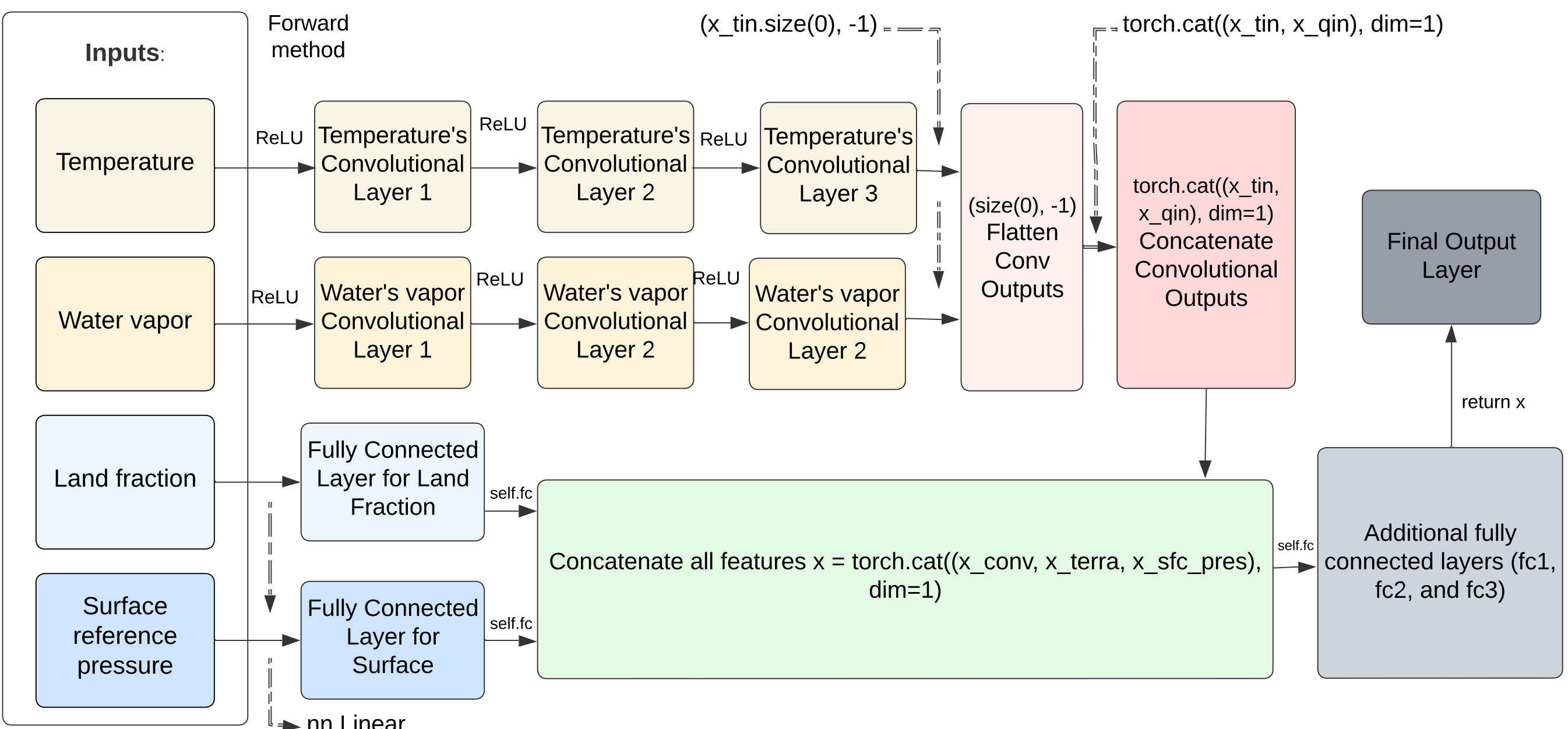
Can a CNN improve climate model performance by enhancing the accuracy of subgrid-scale moist convective tendency predictions?

- One specific aim is to explore the efficacy of convolutional neural network architectures, which contrasts with the current use of fully-connected networks.
- A key aspect of this exploration involves confirming that the CNNs uphold translational invariance, ensuring physical laws are consistent across different altitudes.

Methods



Custom CNN Architecture



- This CNN processes inputs through three sequential convolutional layers for Tin and qin, applies ReLU activations, flattens the outputs, concatenates them with features from fully connected layers applied to terra and sfc_pres, and finally passes the combined features through additional fully connected layers to produce the final output.

Results

- Developed a Convolutional Neural Network (CNN) with multiple convolutional layers to extract hierarchical features from the data.
- Transitioned from fully connected neural networks (FCNNs) to CNNs to better capture and handle spatial data patterns.
- Dropout was proposed in the fully connected layers, leading to a reduction in overfitting and improved model generalization.
- Integration of the CNN into the existing climate model, with necessary adjustments to the training setup for optimal performance (currently in process).

Conclusion

- Once fully integrated, this successful achievement could represent a significant step towards replacing sub-grid routines for moist processes, including convection, in climate models with a more cost-effective and accurate approach.

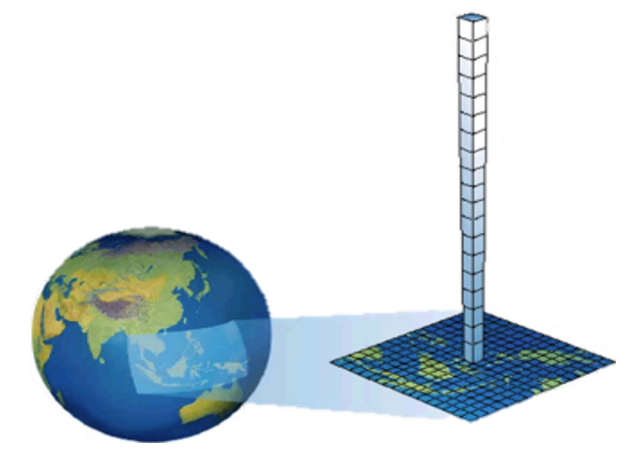


Figure 3: Highlighting the essential grid structure used in our analysis.

This will set a foundation for further research and development, paving the way for replacing outdated sub-grid routines with advanced, machine learning-based methodologies in climate modeling.

Future Work

- Future work includes further refining the CNN through additional hyperparameter tuning, and expanding the model to other climate variables and larger datasets to continuously improve climate modeling accuracy. This also involves exploring its applications in agricultural planning and food security, such as optimizing crop yields and reducing agricultural costs through accurate weather predictions.

Acknowledgments:

This work would not have been possible without the invaluable support and guidance of Griffin Mooers, PhD, and Paul O’Gorman, PhD, who have been instrumental throughout the research process. I would also like to extend my heartfelt thanks to Noelle Wakefield, PhD, Jocelyn Powers, and Carla Garcia for their encouragement in the impeccably organized MSRP program. Lastly, special thanks to the best pod leader, Jesse George-Akpenyi, and Catherine Hazard for their mentorship and contributions.

References:

Yuval, J., O’Gorman, P.A. Stable machine-learning parameterization of subgrid processes for climate modeling at a range of resolutions. *Nat Commun* **11**, 3295 (2020). <https://doi.org/10.1038/s41467-020-17142-3>
Yuval, Janni, et al. “Use of Neural Networks for Stable, Accurate and Physically Consistent Parameterization of Subgrid Atmospheric Processes With Good Performance at Reduced Precision.” *Geophysical Research Letters*, vol. 48, no. 6, 2021, <https://doi.org/10.1029/2020GL091363>.